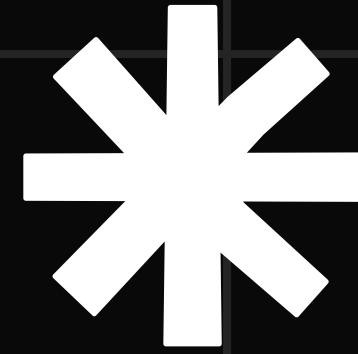


What is....

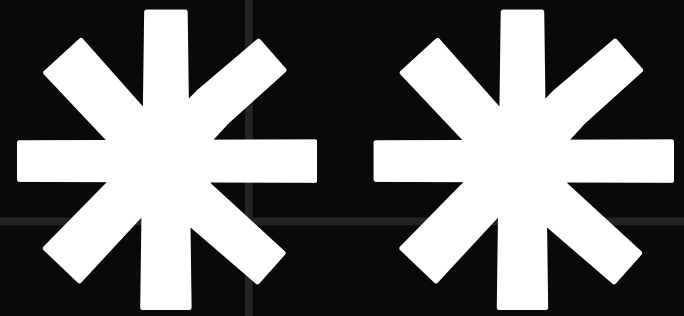


Sign Gesture Detection and Zoom Control

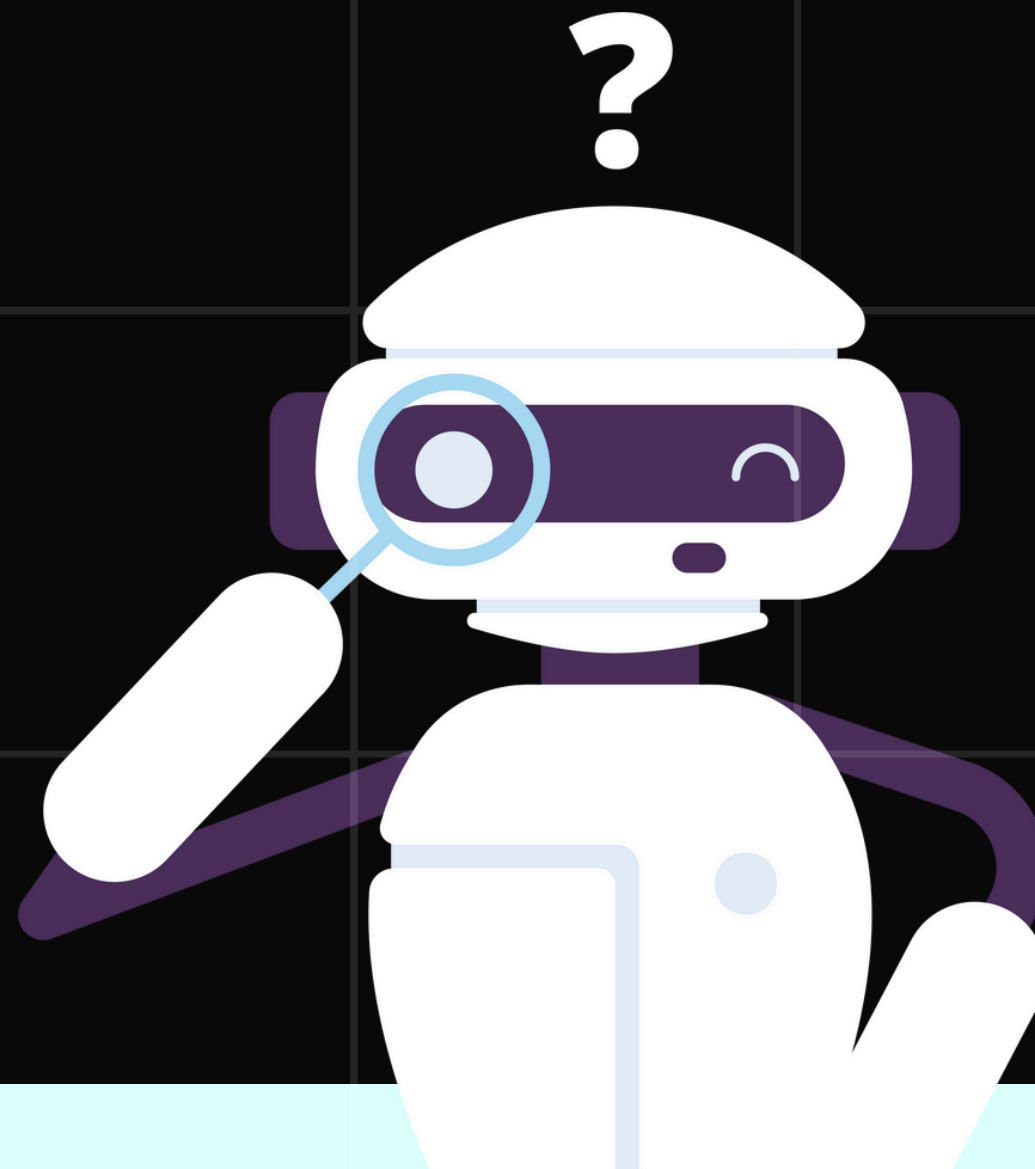
Presented to Purvee Kashyap Mam

Presented by Megha Malviya & Mohini Rathore





Introduction



Objective:

Capture hand gesture images using a webcam.

Libraries used:

OpenCv
Media Pipe
TensorFlow/keras
ImageDataGenerator

Components of AI

Machine Learning

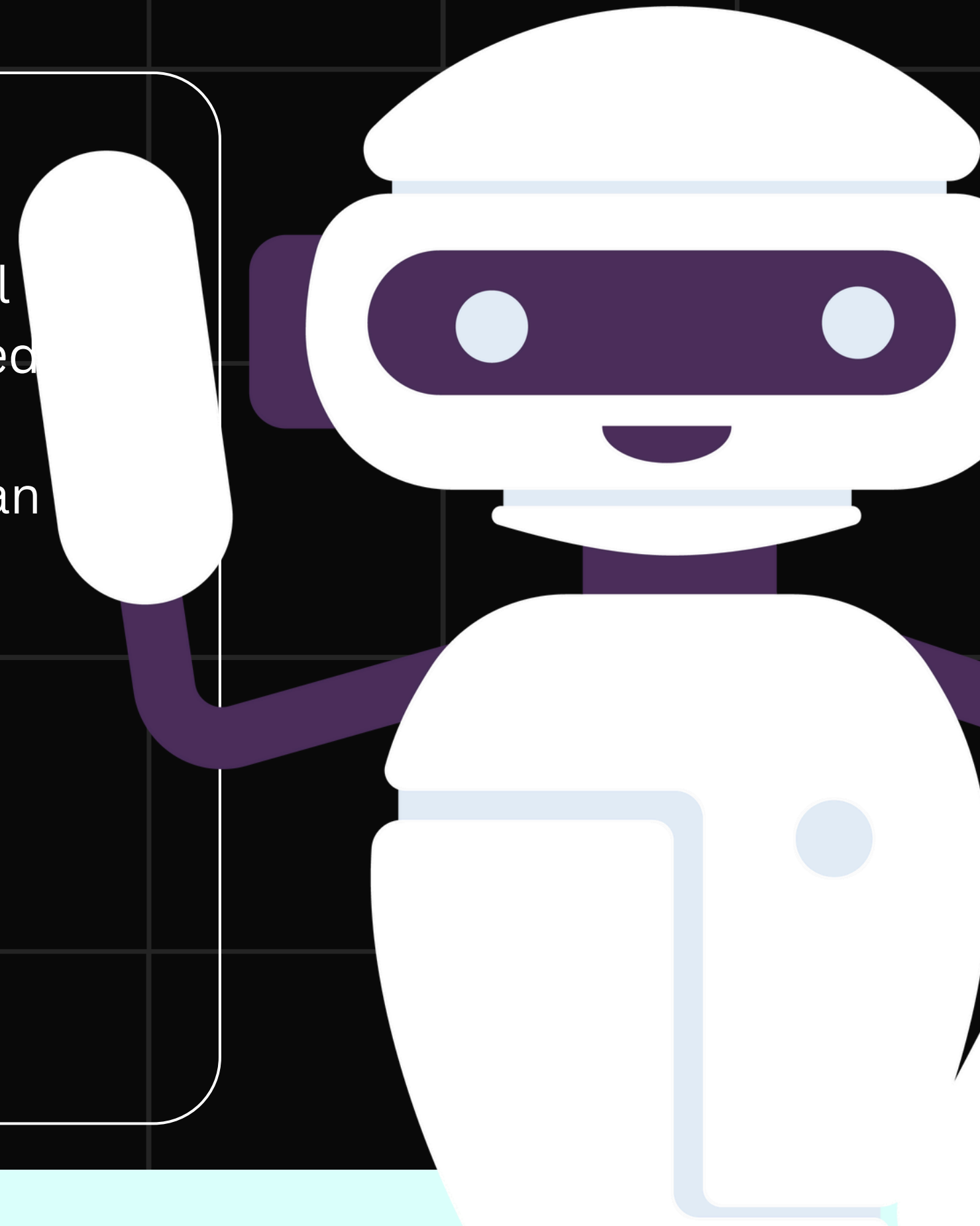
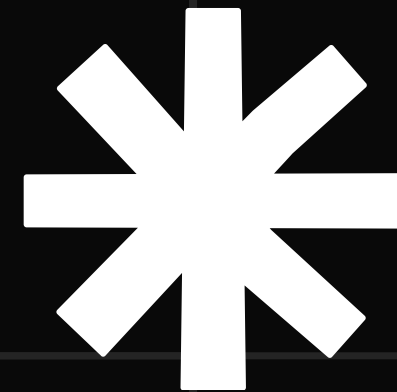
Subset of AI that enables systems to learn from data and improve over time without explicit programming.

NLP, Computer Vision, Robotics

Brief descriptions of other key components contributing to AI's capabilities, including NLP, computer vision, and robotics.

Neural Networks

Introduction to artificial neural networks, inspired by the structure and functioning of the human brain.



AI Applications

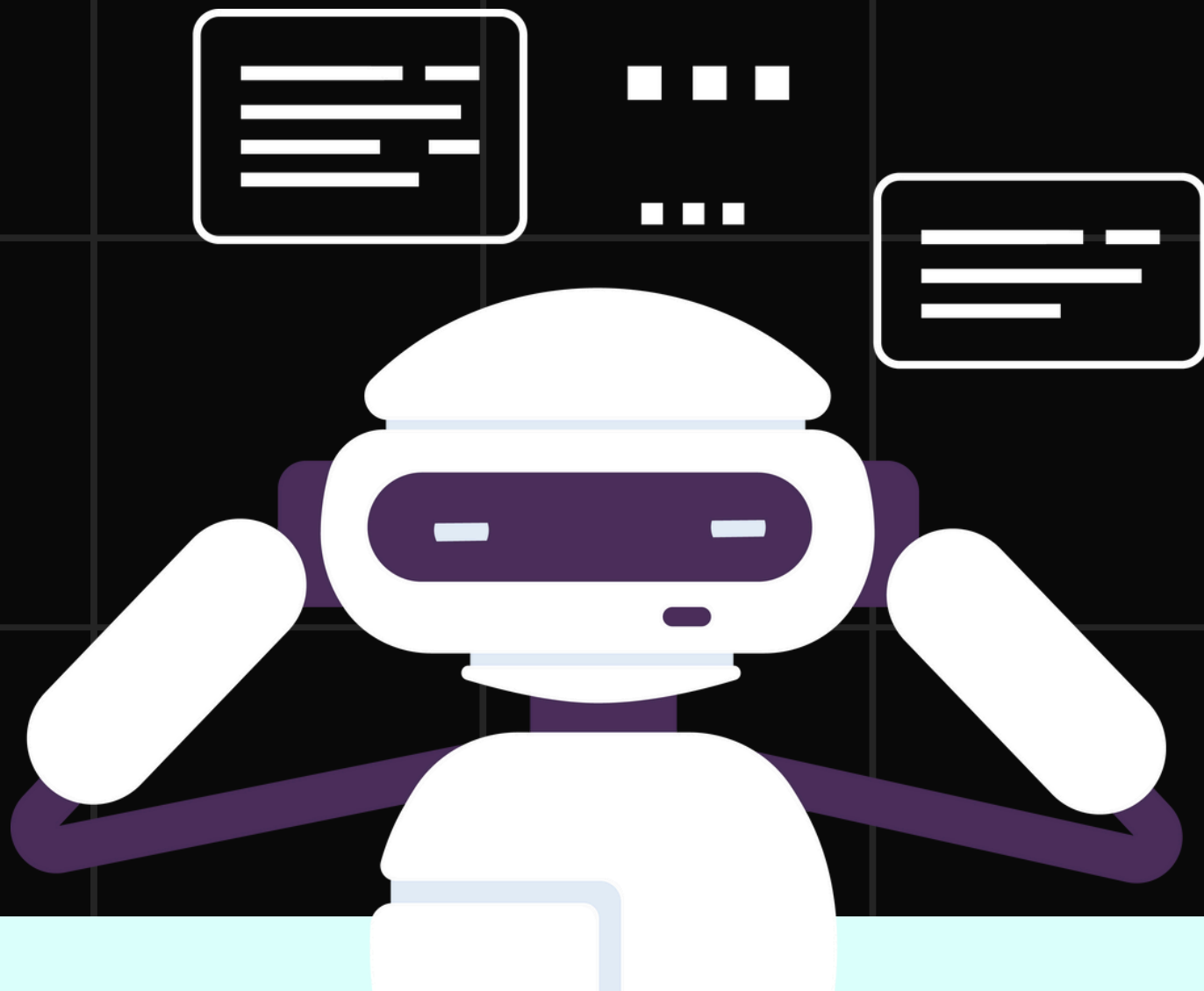
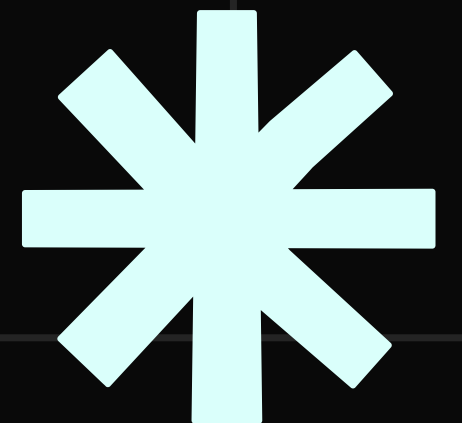
Gesture control
virtual mouse

Gesture based
volume control

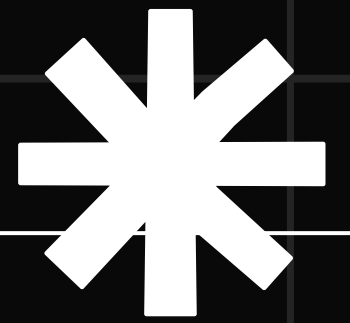
Sign gesture
translator

Gesture based lock
screen

Touchless
drawing
application



Dataset Preparation

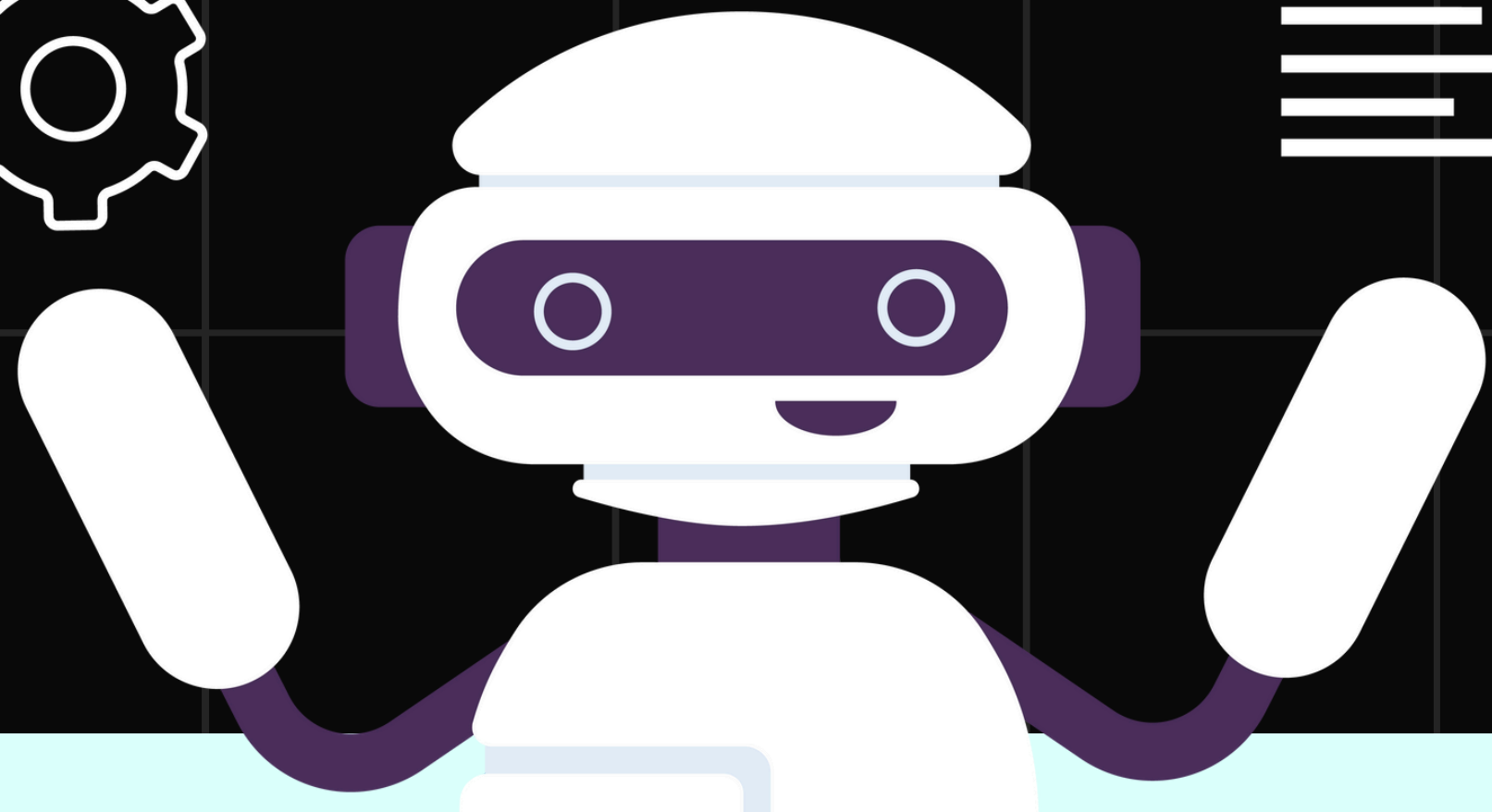
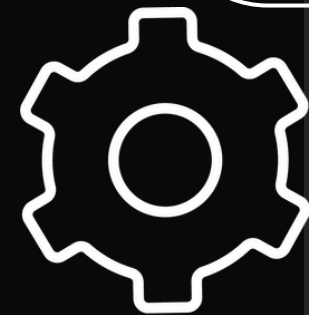
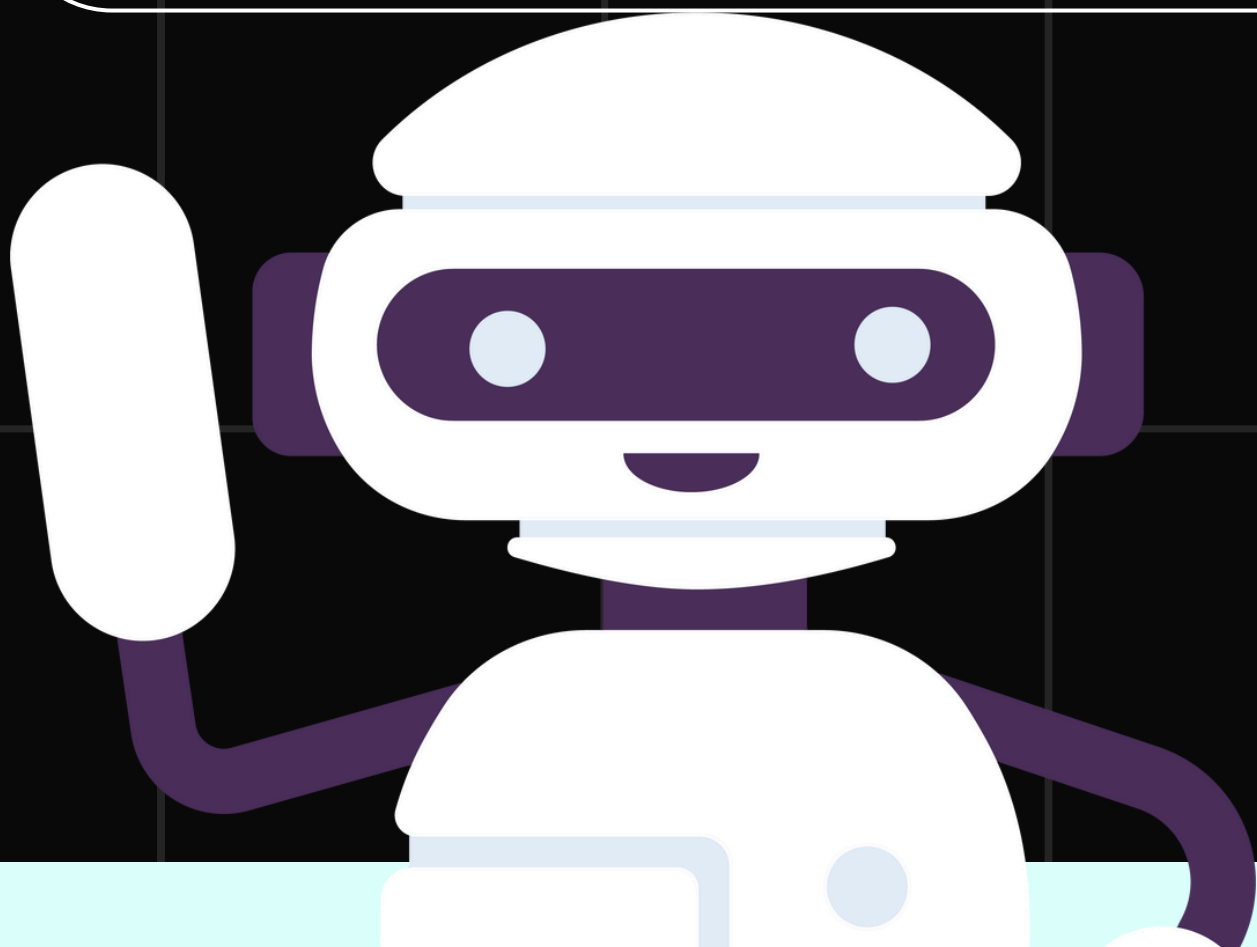


Dataset directory:

gesture_images
Images organized in subfolders
(one per gesture class).

Preprocessing:

Rescaling pixel values
Splitting into training (80%)
and validation (20%)



working

1. Data Collection:

Users capture images for various gestures like thumbs up, peace sign, etc.

2. Model Training:

A CNN model is trained on the collected dataset.

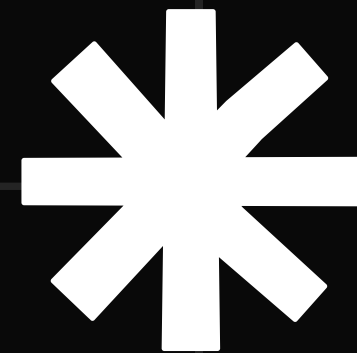
3. Real-Time Detection:

The model predicts gestures live via webcam feed.

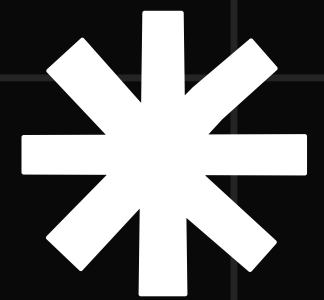
4. Zoom Control:

Mediapipe counts fingers:

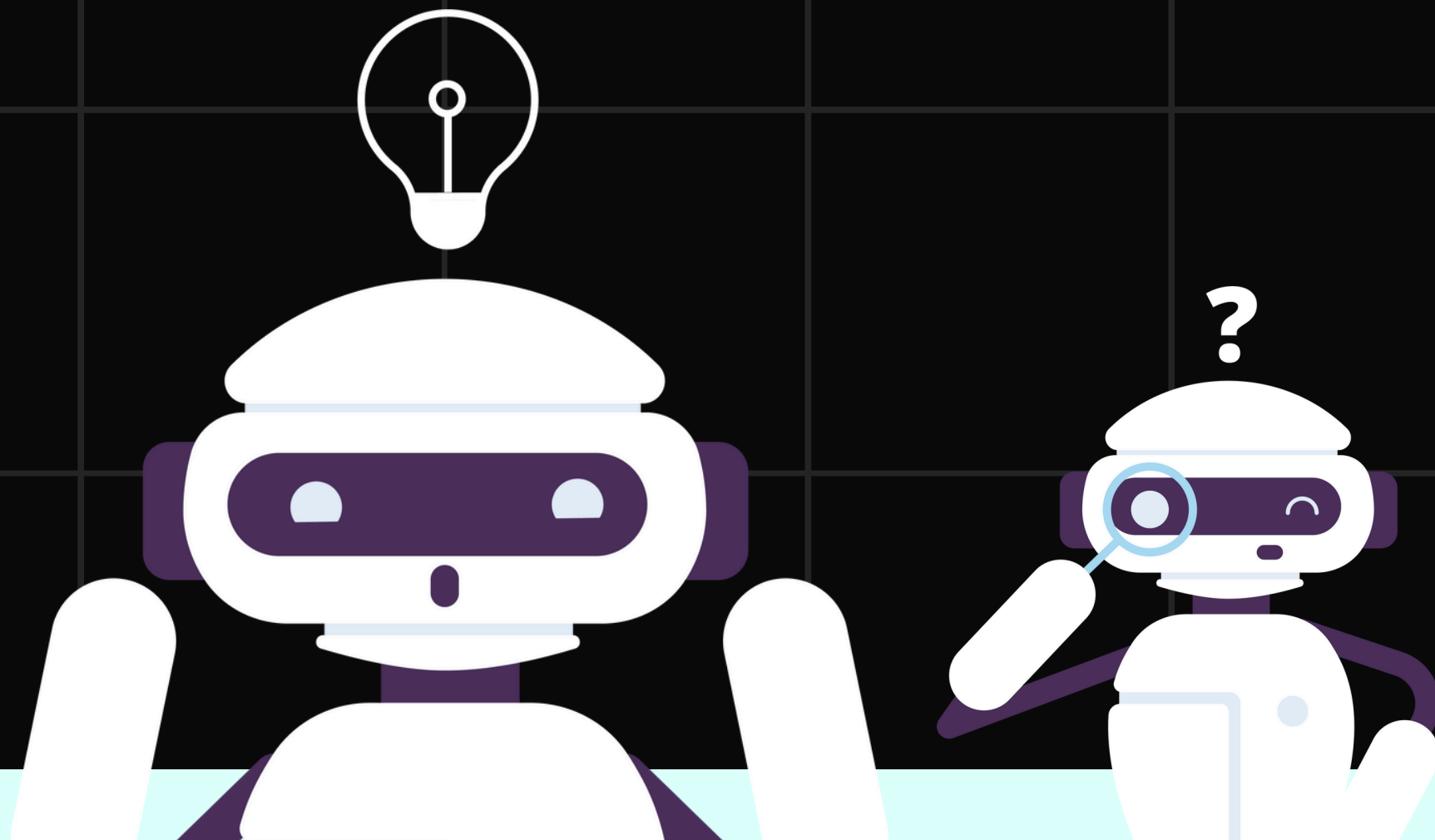
- 4 fingers → Zoom In (Ctrl + +)
- 2 fingers → Zoom Out (Ctrl + -)



```
model = Sequential()  
model.add(Conv2D(32, (3, 3), activation='relu', input_shape=(64, 64, 3)))  
model.add(MaxPooling2D(pool_size=(2, 2)))  
model.add(Conv2D(64, (3, 3), activation='relu'))  
model.add(MaxPooling2D(pool_size=(2, 2)))  
model.add(Flatten())  
model.add(Dense(128, activation='relu'))  
model.add(Dense(len(train_generator.class_indices), activation='softmax'))
```



Model Building Code



Model Training

Training for 10 epochs.

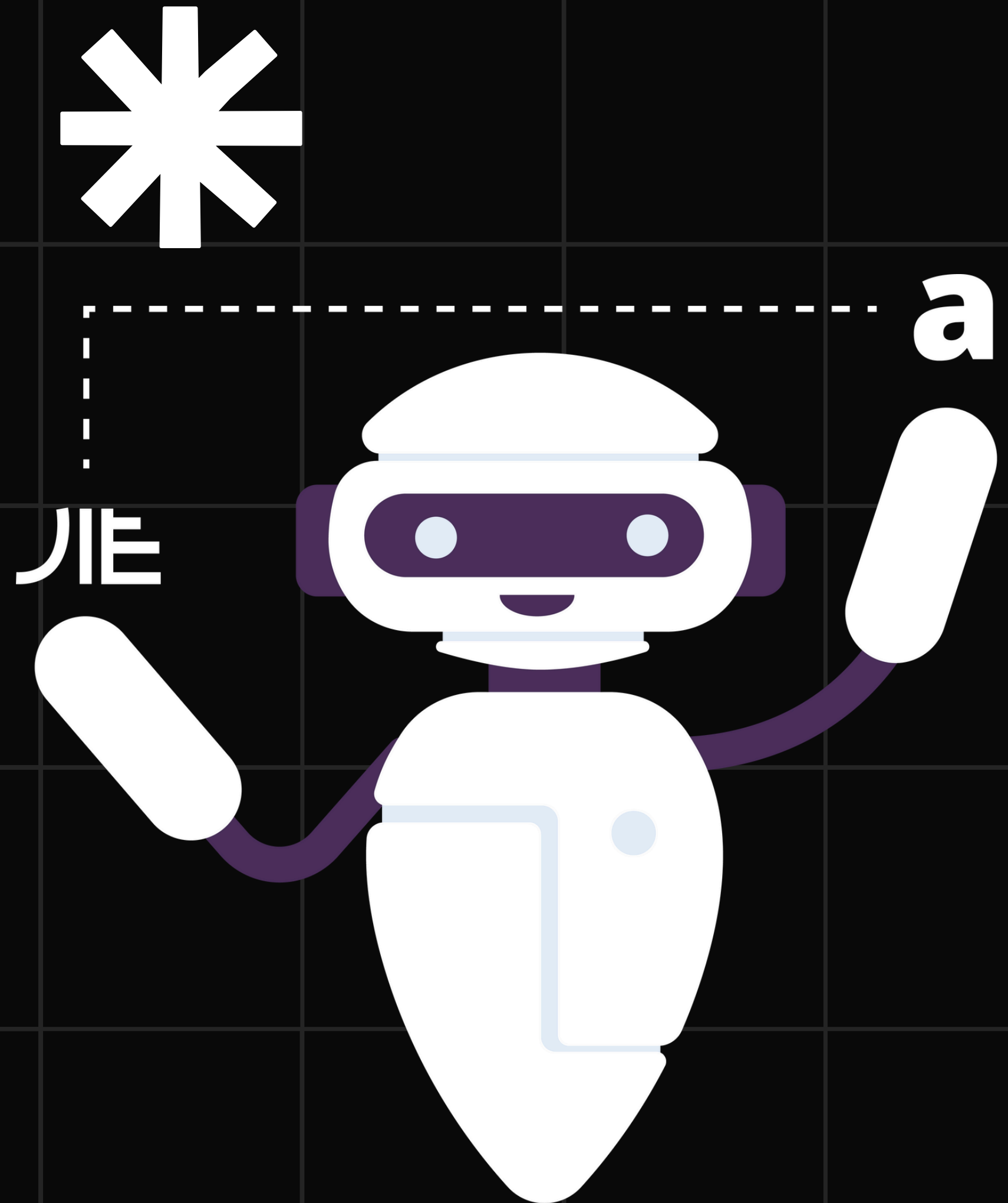
Validation data used to monitor performance.

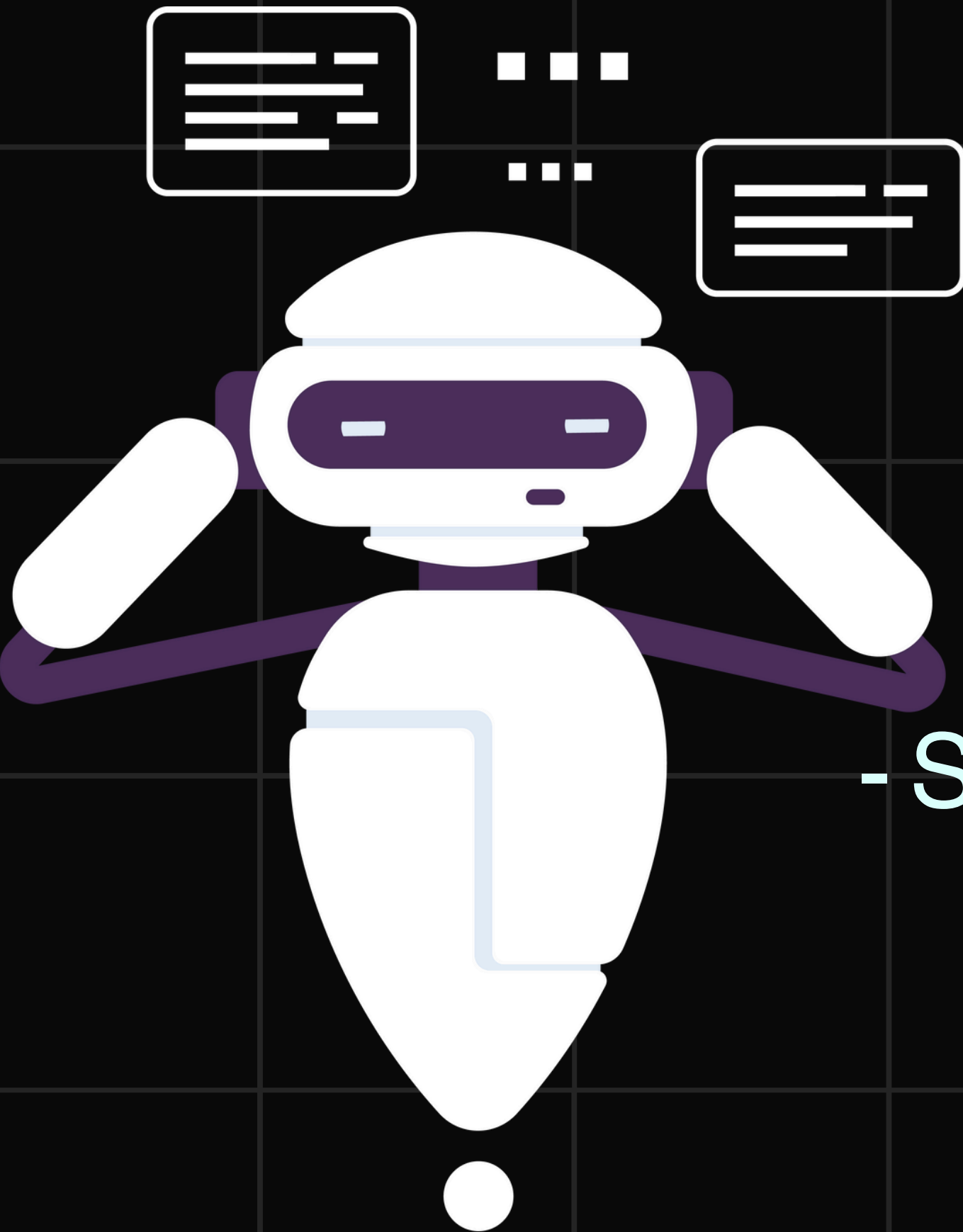
```
model.fit(train_generator,  
validation_data=validation_generator,  
epochs=10)
```

Model Saving

Save model in .h5 format for reuse.

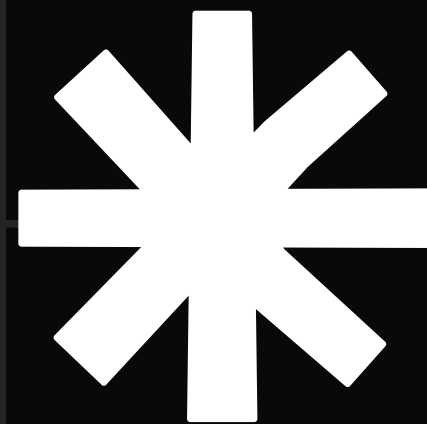
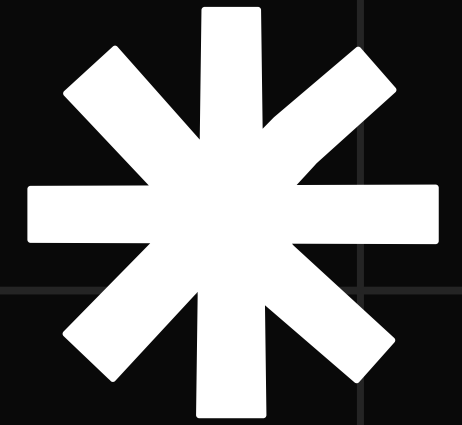
```
model.save('train.model.h5')
```





Results:

- Achieved high real-time accuracy (~90%-95%).
- Smooth gesture recognition at 15-30 FPS.
- Successfully automated zooming through gestures.



Thankyou

@reallygreatsite



